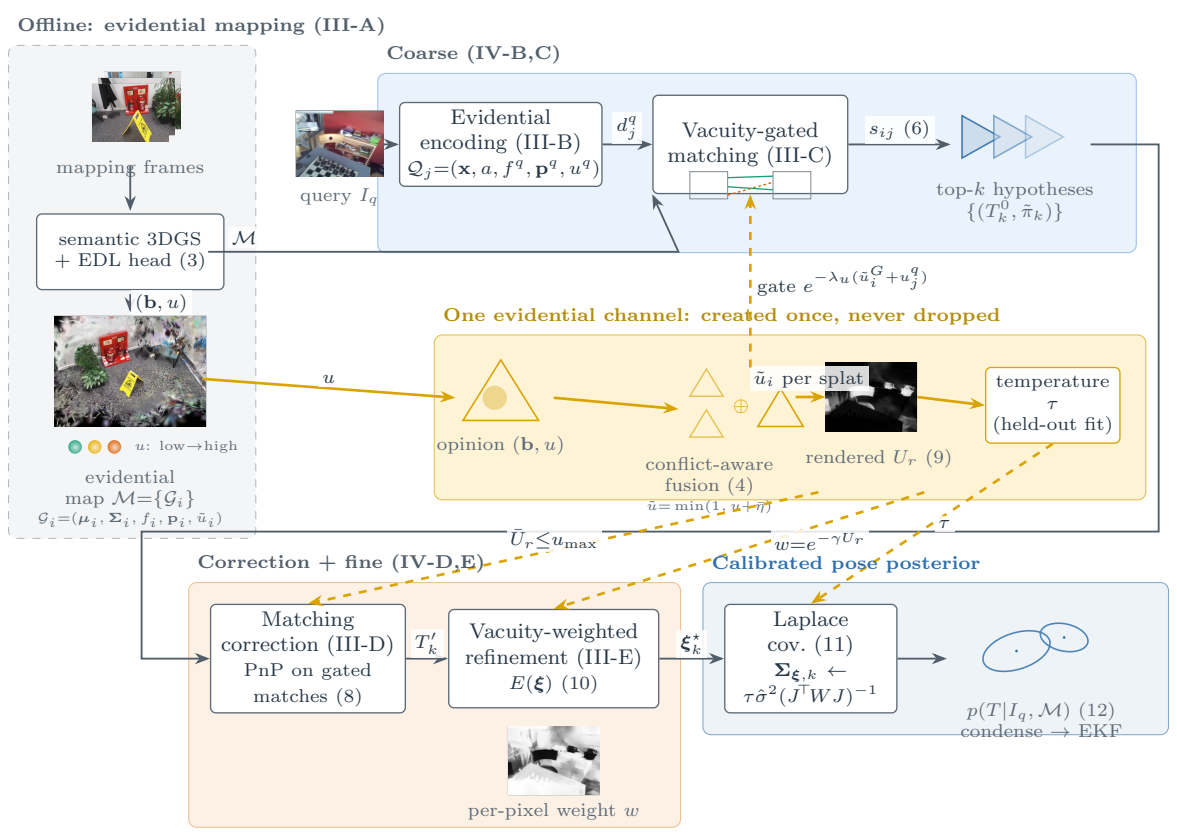

Offline: evidential mapping (III-A)
mapping frames
semantic 3DGS + EDL head (3)
$\mathcal{M}$
$(\mathbf{b}, u)$
u: low→high
evidential map $\mathcal{M}=\{\mathcal{G}_i\}$
$\mathcal{G}_i=(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, f_i, \mathbf{p}_i, \tilde{u}_i)$
Coarse (IV-B,C)
query I_q
Evidential encoding (III-B)
$\mathcal{Q}_j=(\mathbf{x}, a, f^q, \mathbf{p}^q, u^q)$
d_j^q
Vacuity-gated matching (III-C)
s_{ij} (6)
top-k hypotheses
$\{(T_k^0, \tilde{\pi}_k)\}$
gate $e^{-\lambda_u(\tilde{u}_i^G+u_j^q)}$
One evidential channel: created once, never dropped
u
opinion $(\mathbf{b}, u)$
$\tilde{u}_i$ per splat
conflict-aware fusion (4)
$\tilde{u}=\min(1, u+\eta)$
rendered U_r (9)
temperature τ (held-out fit)
$\bar{U}_r \le u_{\max}$
$w=e^{-\gamma U_r}$
τ
Correction + fine (IV-D,E)
Calibrated pose posterior
Matching correction (III-D)
PnP on gated matches (8)
T_k'
Vacuity-weighted refinement (III-E)
$E(\boldsymbol{\xi})$ (10)
$\boldsymbol{\xi}_k^\star$
Laplace cov. (11)
$\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} \leftarrow \tau\hat{\sigma}^2(J^\top W J)^{-1}$
$p(T|I_q, \mathcal{M})$ (12)
condense → EKF
per-pixel weight w